

SafeDreamer: Safe Reinforcement Learning with World Models

24.07.09

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Control and
Intelligent Systems
Laboratory

1. Introduction

(2) Contributions

- (OSRP) **Online planning** within the world model to **satisfy constraints** by using **Constrained Cross-Entropy Method** in the planning process.

Method	Vision Input	Low-dimensional Input	Lagrangian-based	Online Planning	Background Planning
LAMBDA (As et al., 2022)	✓	X	✓	X	✓
SafeDreamer (ours)	✓	✓	✓	✓	✓

- integrate Lagrangian methods with the safety-reward online (OSRP-Lag) and background (BSRP-Lag) planning within the world model to **balance long-term reward and cost**.
- SafeDreamer achieving nearly **zero-cost performance** in the Safety-Gymnasium benchmark.

2. Problem formulation

[Safe Model-based RL Problem]

$$\max_{\pi_{\theta} \in \Pi_{\theta}} J_{\phi}^R(\pi_{\theta}) \quad \text{s.t.} \quad J_{\phi}^C(\pi_{\theta}) \leq \underbrace{b}_{\text{threshold}}, \text{ where} \quad (2)$$

$$J_{\phi}^R(\pi_{\theta}) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_{t+1} | s_t, a_t) \mid s_0 \sim \mu, \underbrace{s_{t+1} \sim \mathbb{P}_{\phi}(\cdot | s_t, a_t)}_{\text{parameterized world model}}, a_t \sim \pi_{\theta} \right], \quad (3)$$

$$J_{\phi}^C(\pi_{\theta}) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t C(s_{t+1} | s_t, a_t) \mid s_0 \sim \mu, s_{t+1} \sim \mathbb{P}_{\phi}(\cdot | s_t, a_t), a_t \sim \pi_{\theta} \right]. \quad (4)$$

2. Problem formulation

[Safe Model-based RL Problem]

$$\max_{\pi_{\theta} \in \Pi_{\theta}} J_{\phi}^R(\pi_{\theta}) \quad \text{s.t.} \quad J_{\phi}^C(\pi_{\theta}) \leq \underline{b}, \quad \text{where} \quad \text{\textit{threshold}} \quad (2)$$

$$J_{\phi}^R(\pi_{\theta}) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_{t+1} | s_t, a_t) \mid s_0 \sim \mu, \underbrace{s_{t+1} \sim \mathbb{P}_{\phi}(\cdot | s_t, a_t)}_{\text{\textit{parameterized world model}}}, a_t \sim \pi_{\theta} \right], \quad (3)$$

$$J_{\phi}^C(\pi_{\theta}) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t C(s_{t+1} | s_t, a_t) \mid s_0 \sim \mu, s_{t+1} \sim \mathbb{P}_{\phi}(\cdot | s_t, a_t), a_t \sim \pi_{\theta} \right]. \quad (4)$$



$$\max_{\pi_{\theta} \in \Pi_{\theta}} \min_{\lambda_p \geq 0} J_{\phi}^R(\pi_{\theta}) - \lambda_p (J_{\phi}^C(\pi_{\theta}) - b). \quad (5)$$

* The Lagrangian method stands as a general solution for SafeRL, and the commonly used ones are Augmented Lagrangian (Dai et al., 2023) and PID Lagrangian (Stooke et al., 2020). However, it has been observed that employing Lagrangian approaches for model-free SafeRL results in suboptimal performance under low-cost threshold settings, attributable to inaccuracies in the critic's estimation.

3. Methods

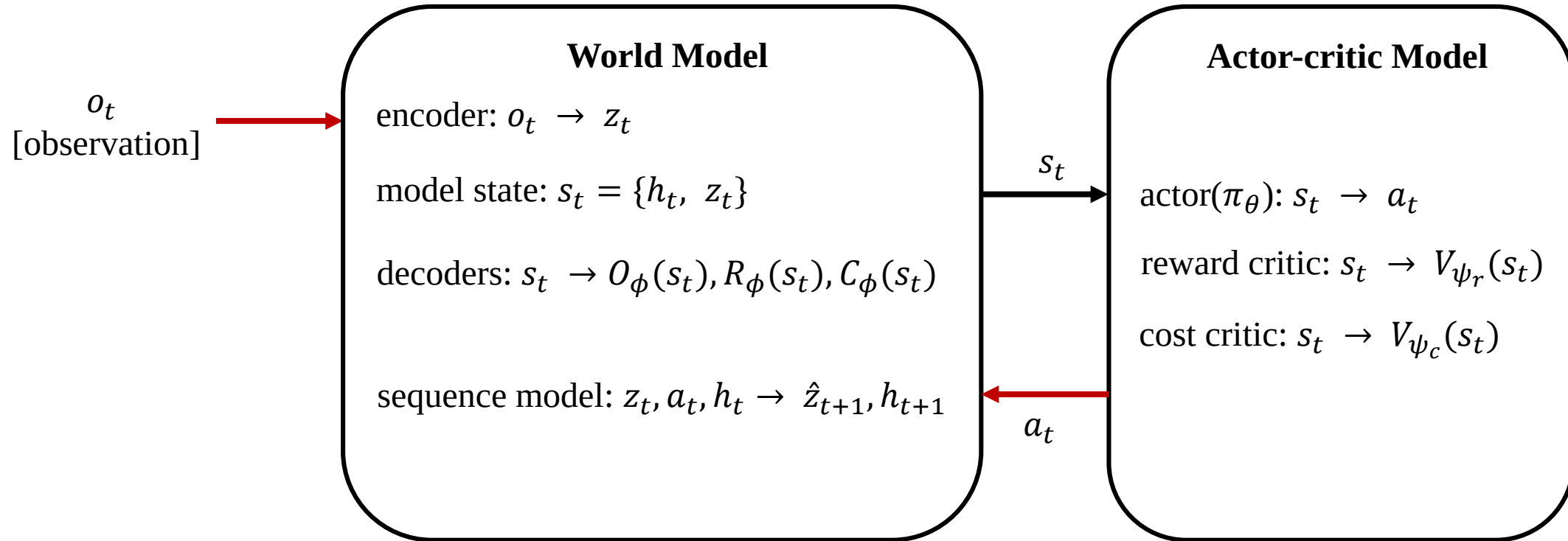
(1) Online planning vs Offline planning

Feature	Online Planning	Background Planning
Definition	Real-time planning during interaction	Offline planning using simulations
Execution	At each decision step	During non-interactive periods
Usage	Immediate decision-making	Policy refinement and training
Advantages	Responsive, adaptive	Computationally and sample efficient
Disadvantages	High computational cost, less efficient	Less adaptive, not real-time
SafeDreamer Example	OSRP-Lag	BSRP-Lag

* produce imagined latent rollouts of length $T = 15$

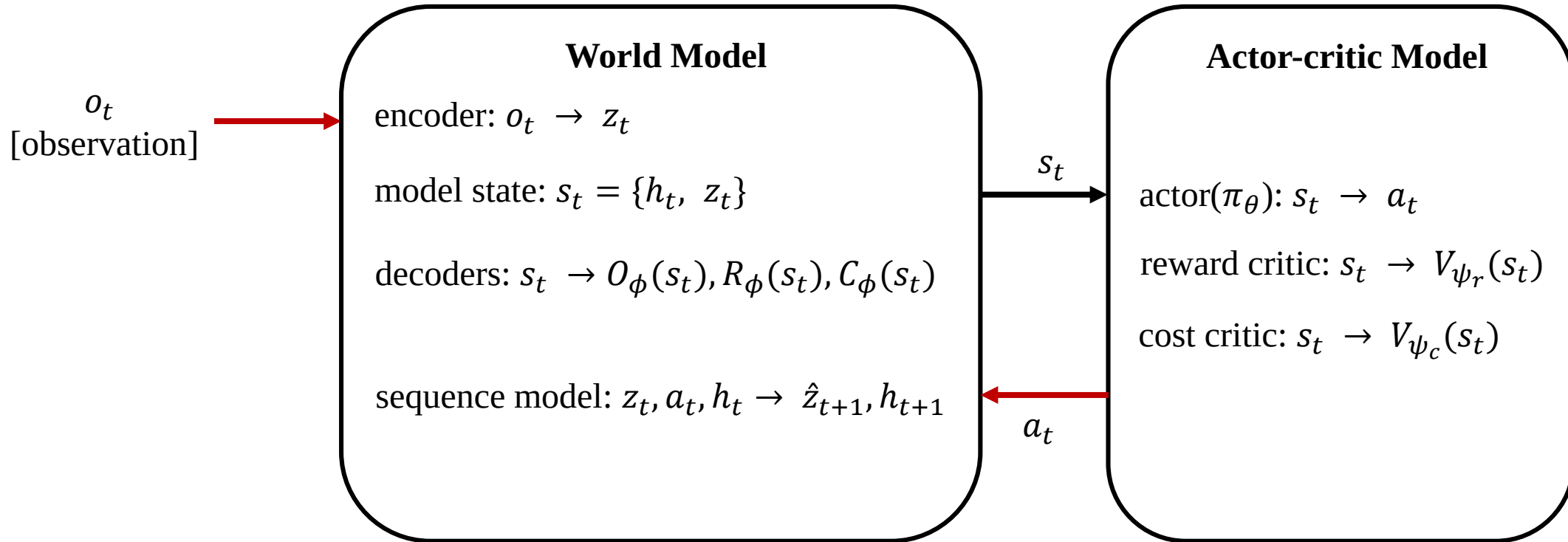
3. Methods

(2) Model Components



3. Methods

(2) Model Components



[trajectories planning by π_θ]

$$\{s_i, a_i, s_{i+1}\}_{i=t}^{t + \text{planning horizon}} + R_\phi(s_i), C_\phi(s_i), V_{\psi_r}(s_i), V_{\psi_c}(s_i)$$

3. Methods

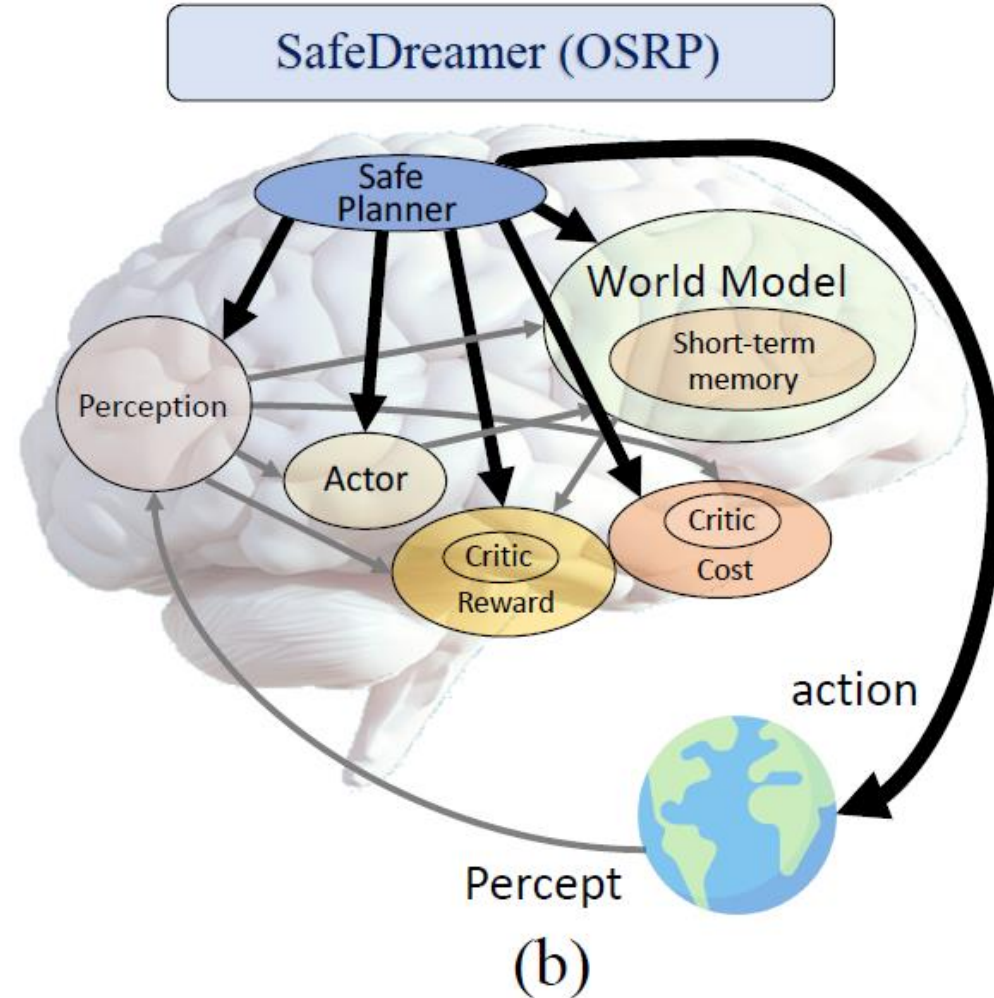
(2) Online Safety-Reward Planning(OSRP) via World Model

Algorithm 1: Online Safety-Reward Planning.

```

1 Input: current model state  $s_t$ , planning horizon  $H$ ,
2 num sample/policy/safe trajectories  $N_{\pi_N}, N_{\pi_\theta}, N_s$ ,
3 Lagrangian multiplier  $\lambda_p$ , cost limit  $b, \mu^0, \sigma^0$  for  $\mathcal{N}$ 
4 for  $j \leftarrow 1$  to  $J$  do
5   Init. empty safe/candidate/elite actions set
6    $A_s, A_c, A_e$ 
7   Sample  $N_{\pi_N} + N_{\pi_\theta}$  traj.  $\{s_i, a_i, s_{i+1}\}_{i=t}^{t+H}$ 
8   using  $\mathcal{N}(\mu^{j-1}, (\sigma^{j-1})^2 \mathbf{I}), \pi_\theta$  within  $S_\phi$  with
9    $s_t$  as the initial state
10  for all  $N_{\pi_N} + N_{\pi_\theta}$  trajectories do
11    Init. trajectory cost  $J_\phi^{C,H}(\pi) \leftarrow 0$ 
12    for  $t \leftarrow H$  to  $1$  do
13      Compute  $R^\lambda(s_t), C^\lambda(s_t)$  via
14      Equation (6)
15       $J_\phi^{C,H} \leftarrow \gamma J_\phi^{C,H} + C_\phi(s_t)$ 
16    end
17     $J_\phi^R, J_\phi^C \leftarrow R^\lambda(s_1), C^\lambda(s_1)$ 
18     $J_\phi^{C'} \leftarrow (J_\phi^{C,H} / H)L$ 
19    if  $J_\phi^{C'} < b$  then
20       $A_s \leftarrow A_s \cup \{a_{t:t+H}\}$ 
21    end
22  end
23  Select sorting key  $\Omega$ , candidate action set  $A_c$ 
24  Select the top-k action trajectories with highest
25   $\Omega$  values among  $A_c$  as elite actions  $A_e$ 
26   $\mu^j, \sigma^j = \text{MEAN}(A_e), \text{STD}(A_e)$ 
27 Return:  $a \sim \mathcal{N}(\mu^J, (\sigma^J)^2 \mathbf{I})$ 

```



3. Methods

(2) Online Safety-Reward Planning(OSRP) via World Model

CCEM STEP 1: add safe action set A_s

[trajectories planning by π_θ]

$$\{s_i, a_i, s_{i+1}\}_{i=t}^{t + \text{planning horizon}} + R_\phi(s_i), C_\phi(s_i), V_{\psi_r}(s_i), V_{\psi_c}(s_i)$$

[trajectories planning by $\pi_{\mathcal{N}}$]

$$\{s_i, a_i, s_{i+1}\}_{i=t}^{t + \text{planning horizon}} \text{ with } a_i \sim \mathcal{N}(\mu^{j-1}, (\sigma^{j-1})^2 I) \text{ and sequence model } S_\phi$$

3. Methods

(2) Online Safety-Reward Planning(OSRP) via World Model

CCEM STEP 1: add safe action set A_s

for all $N_{\pi_{\mathcal{N}}} + N_{\pi_{\theta}}$ trajectories **do**

Init. trajectory cost $J_{\phi}^{C,H}(\pi) \leftarrow 0$

for $t \leftarrow H$ **to** 1 **do**

Compute $R^{\lambda}(s_t), C^{\lambda}(s_t)$ via
Equation (6)

$J_{\phi}^{C,H} \leftarrow \gamma J_{\phi}^{C,H} + C_{\phi}(s_t)$

end

$J_{\phi}^R, J_{\phi}^C \leftarrow R^{\lambda}(s_1), C^{\lambda}(s_1)$

$J_{\phi}^{C'} \leftarrow (J_{\phi}^{C,H} / H)L$

if $J_{\phi}^{C'} < b$ **then**

$A_s \leftarrow A_s \cup \{a_{t:t+H}\}$

end

end

실제 $state, action$ 에 따라 얻는 $reward/cost$
 $\approx$ output of cost model

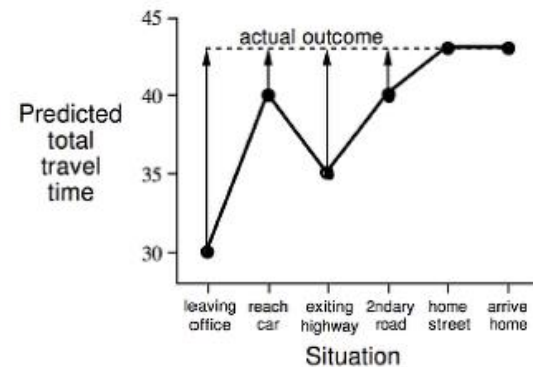
3. Methods

(2) Online Safety-Reward Planning(OSRP) via World Model

CCEM STEP 1: add safe action set A_s

[TD(0)]

$$V(S_t) \leftarrow V(S_t) + \alpha (R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$

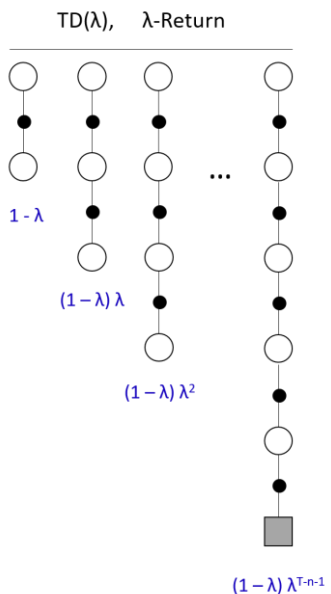


3. Methods

(2) Online Safety-Reward Planning(OSRP) via World Model

CCEM STEP 1: add safe action set A_s

[TD(λ)]



$$C^\lambda(s_t) = \begin{cases} \frac{C_\phi(s_t) + \gamma ((1 - \lambda) \underbrace{V_{\psi_c}(s_{t+1})}_{\text{value from cost critic}} + \lambda C^\lambda(s_{t+1}))}{\underbrace{V_{\psi_c}(s_t)}} & \text{if } t < H \\ V_{\psi_c}(s_t) & \text{if } t = H \end{cases} \quad (6)$$

3. Methods

(2) Online Safety-Reward Planning(OSRP) via World Model

CCEM STEP 1: add safe action set A_s

```
for all  $N_{\pi_{\mathcal{N}}} + N_{\pi_{\theta}}$  trajectories do  
  Init. trajectory cost  $J_{\phi}^{C,H}(\pi) \leftarrow 0$   
  for  $t \leftarrow H$  to 1 do  
    Compute  $R^{\lambda}(s_t), C^{\lambda}(s_t)$  via  
    Equation (6)  
     $J_{\phi}^{C,H} \leftarrow \gamma J_{\phi}^{C,H} + C_{\phi}(s_t)$   
  end  
   $J_{\phi}^R, J_{\phi}^C \leftarrow R^{\lambda}(s_1), C^{\lambda}(s_1)$   
   $J_{\phi}^{C'} \leftarrow (J_{\phi}^{C,H} / H)L$   
  if  $J_{\phi}^{C'} < b$  then  
     $A_s \leftarrow A_s \cup \{a_{t:t+H}\}$   
  end  
end
```

3. Methods

(2) Online Safety-Reward Planning(OSRP) via World Model

CCEM STEP 1: add safe action set A_s

```
for all  $N_{\pi_{\mathcal{N}}} + N_{\pi_{\theta}}$  trajectories do  
  Init. trajectory cost  $J_{\phi}^{C,H}(\pi) \leftarrow 0$   
  for  $t \leftarrow H$  to 1 do  
    Compute  $R^{\lambda}(s_t), C^{\lambda}(s_t)$  via  
    Equation (6)  
     $J_{\phi}^{C,H} \leftarrow \gamma J_{\phi}^{C,H} + C_{\phi}(s_t)$   
  end  
   $J_{\phi}^R, J_{\phi}^C \leftarrow R^{\lambda}(s_1), C^{\lambda}(s_1)$   
   $J_{\phi}^{C'} \leftarrow (J_{\phi}^{C,H} / H)L$   
  if  $J_{\phi}^{C'} < b$  then  
     $A_s \leftarrow A_s \cup \{a_{t:t+H}\}$   
  end  
end
```

$$J_{\phi}^{C,H} = \sum_t^{t+H} \gamma^t C_{\phi}(s_t)$$

total cost over horizon H (computed with cost model)



$$J_{\phi}^{C'} = (J_{\phi}^{C,H} / H)L$$

alternative estimation for avoiding error of cost critic

3. Methods

(2) Online Safety-Reward Planning(OSRP) via World Model

CCEM STEP 1: add safe action set A_s

```
for all  $N_{\pi_{\mathcal{N}}} + N_{\pi_{\theta}}$  trajectories do
  Init. trajectory cost  $J_{\phi}^{C,H}(\pi) \leftarrow 0$ 
  for  $t \leftarrow H$  to  $1$  do
    Compute  $R^{\lambda}(s_t), C^{\lambda}(s_t)$  via
      Equation (6)
     $J_{\phi}^{C,H} \leftarrow \gamma J_{\phi}^{C,H} + C_{\phi}(s_t)$ 
  end
   $J_{\phi}^R, J_{\phi}^C \leftarrow R^{\lambda}(s_1), C^{\lambda}(s_1)$ 
   $J_{\phi}^{C'} \leftarrow (J_{\phi}^{C,H} / H)L$ 
  if  $J_{\phi}^{C'} < b$  then
     $A_s \leftarrow A_s \cup \{a_{t:t+H}\}$ 
  end
end
end
```

3. Methods

(2) Online Safety-Reward Planning(OSRP) via World Model

CCEM STEP 2: set candidate action set A_c and select elite action set A_e

```
    Select sorting key  $\Omega$ , candidate action set  $A_c$   
    Select the top-k action trajectories with highest  
     $\Omega$  values among  $A_c$  as elite actions  $A_e$   
     $\mu^j, \sigma^j = \text{MEAN}(A_e), \text{STD}(A_e)$ 
```

end

Return: $a \sim \mathcal{N}(\mu^J, (\sigma^J)^2 \mathbf{I})$

3. Methods

(2) Online Safety-Reward Planning(OSRP) via World Model

CCEM STEP 2: set candidate action set A_c and select elite action set A_e

Select sorting key Ω , candidate action set A_c
Select the top-k action trajectories with highest
 Ω values among A_c as elite actions A_e

$\mu^j, \sigma^j = \text{MEAN}(A_e), \text{STD}(A_e)$

end

Return: $a \sim \mathcal{N}(\mu^J, (\sigma^J)^2 \mathbf{I})$

N_s : num of safe trajectories (user input)

If $|A_s| < N_s$

$A_c = \text{all trajectories}, \quad \text{sorting key } \Omega = -J_{\phi}^{C'}$

If $|A_s| \geq N_s$

$A_c = A_s, \quad \text{sorting key } \Omega = J_{\phi}^R - \lambda_p J_{\phi}^C$

3. Methods

(2) Online Safety-Reward Planning(OSRP) via World Model

CCEM STEP 2: set candidate action set A_c and select elite action set A_e

```
| Select sorting key  $\Omega$ , candidate action set  $A_c$   
| Select the top-k action trajectories with highest  
|  $\Omega$  values among  $A_c$  as elite actions  $A_e$   
|  $\mu^j, \sigma^j = \text{MEAN}(A_e), \text{STD}(A_e)$   
end  
Return:  $a \sim \mathcal{N}(\mu^J, (\sigma^J)^2 \mathbf{I})$ 
```

$$\mu^j = \frac{1}{k} \sum_{i=1}^k A_e^i, \sigma^j = \sqrt{\frac{1}{k} \sum_{i=1}^k (A_e^i - \mu^j)^2}$$

update action distribution

3. Methods

(2) Online Safety-Reward Planning(OSRP) via World Model

CCEM STEP 3: return after J iterations.

```
|  Select sorting key  $\Omega$ , candidate action set  $A_c$   
|  Select the top-k action trajectories with highest  
|     $\Omega$  values among  $A_c$  as elite actions  $A_e$   
|   $\mu^j, \sigma^j = \text{MEAN}(A_e), \text{STD}(A_e)$   
end  
Return:  $a \sim \mathcal{N}(\mu^J, (\sigma^J)^2 \mathbf{I})$ 
```

3. Methods

(3) Online Safety-Reward Planning(OSRP) - Lag

output(estimation) of cost model (world model) $\neq$ real episode cost

$$J_{\phi}^R - \lambda_p J_{\phi}^C$$

To balance long-term reward and cost:

Algorithm 3: PID Lagrangian (Stooke et al., 2020).

Input: Proportional coefficient K_p , integral coefficient K_i , differential coefficient K_d

1 Initialize previous integral item: $I^0 \leftarrow 0$;

2 Initialize previous episode cost: $J_C^0 \leftarrow 0$;

3 **while** iteration k continues **do**

4 Receive cost J_C^k ;

5 $P \leftarrow J_C^k - d$;

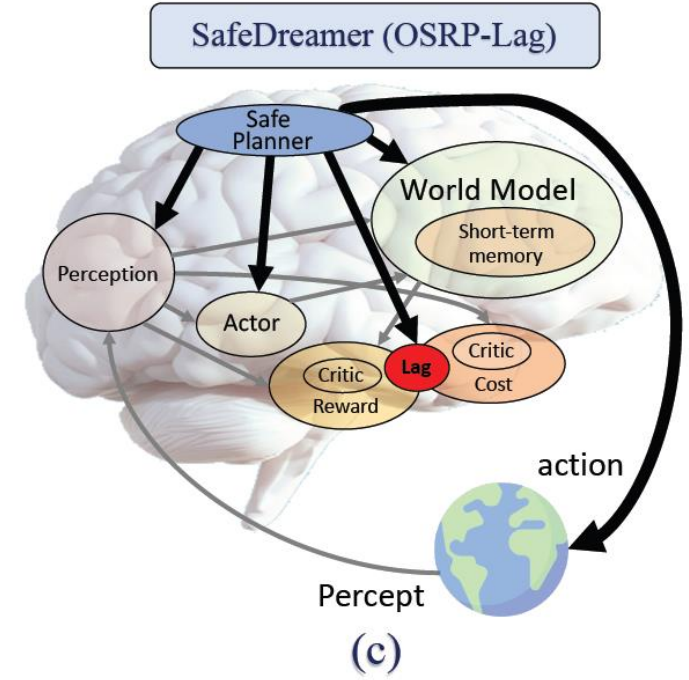
6 $D \leftarrow \max(0, J_C^k - J_C^{k-1})$;

7 $I^k \leftarrow \max(0, I^{k-1} + D)$;

8 $\lambda_p \leftarrow \max(0, K_p P + K_i I^k + K_d D)$;

9 Return Lagrangian multiplier λ_p ;

10 **end**



* The safety planner might fail to ensure zero-cost when the required hazard detection horizon surpasses the planning horizon. The process for updating the Lagrangian multiplier remains consistent with PID Lagrangian method and depends on the episode cost encountered during the interaction between agent and environment.

3. Methods

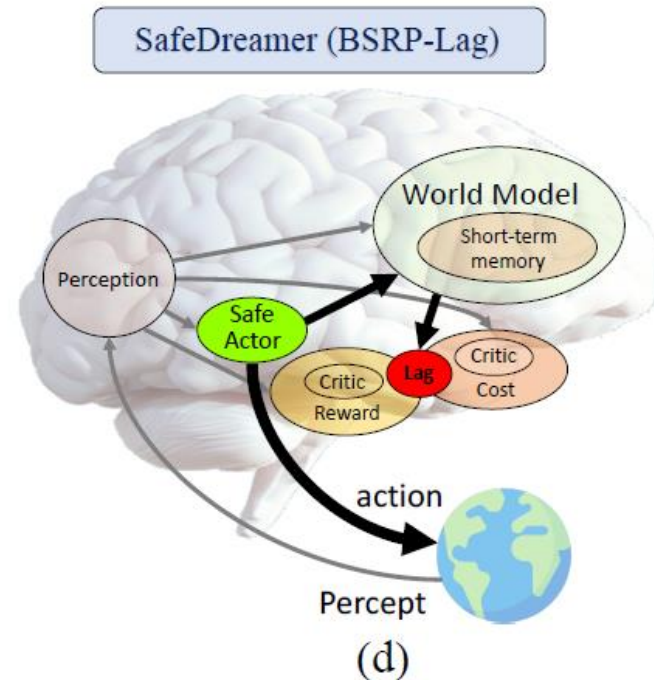
(4) Background Safety-Reward Planning(OSRP) - Lag

(a) produce imagined latent rollouts of length $T = 15$ within the world model. <차이점>

(b) begin with observations from the replay buffer, sampling actions from the actor, and observations from the world model.

(c) world model also predict rewards and costs, from which we compute $TD(\lambda)$ value.

똑같이 PID lagrangian method를 사용하지만, cost model output을 이용함.
환경과 직접 interact하며 실제 cost를 반영하는 OSRP-Lag에서와 다르게, cost model error가 포함된다는 단점이 있음.



4. EXPERIMENTAL RESULTS

(1) Metrics

- Average undiscounted reward return over E episodes: $\hat{J}(\pi) = \frac{1}{E} \sum_{i=1}^E \sum_{t=0}^{T_{\text{ep}}} r_t$.
- Average undiscounted cost return over E episodes: $\hat{J}_c(\pi) = \frac{1}{E} \sum_{i=1}^E \sum_{t=0}^{T_{\text{ep}}} c_t$.
- Average cost throughout the entire training phase, namely the *cost rate*: Given a total of T interaction steps, we define the cost rate $\rho_c(\pi) = \frac{\sum_{t=0}^T c_t}{T}$.

4. EXPERIMENTAL RESULTS

(2) Safety-Gymnasium tasks

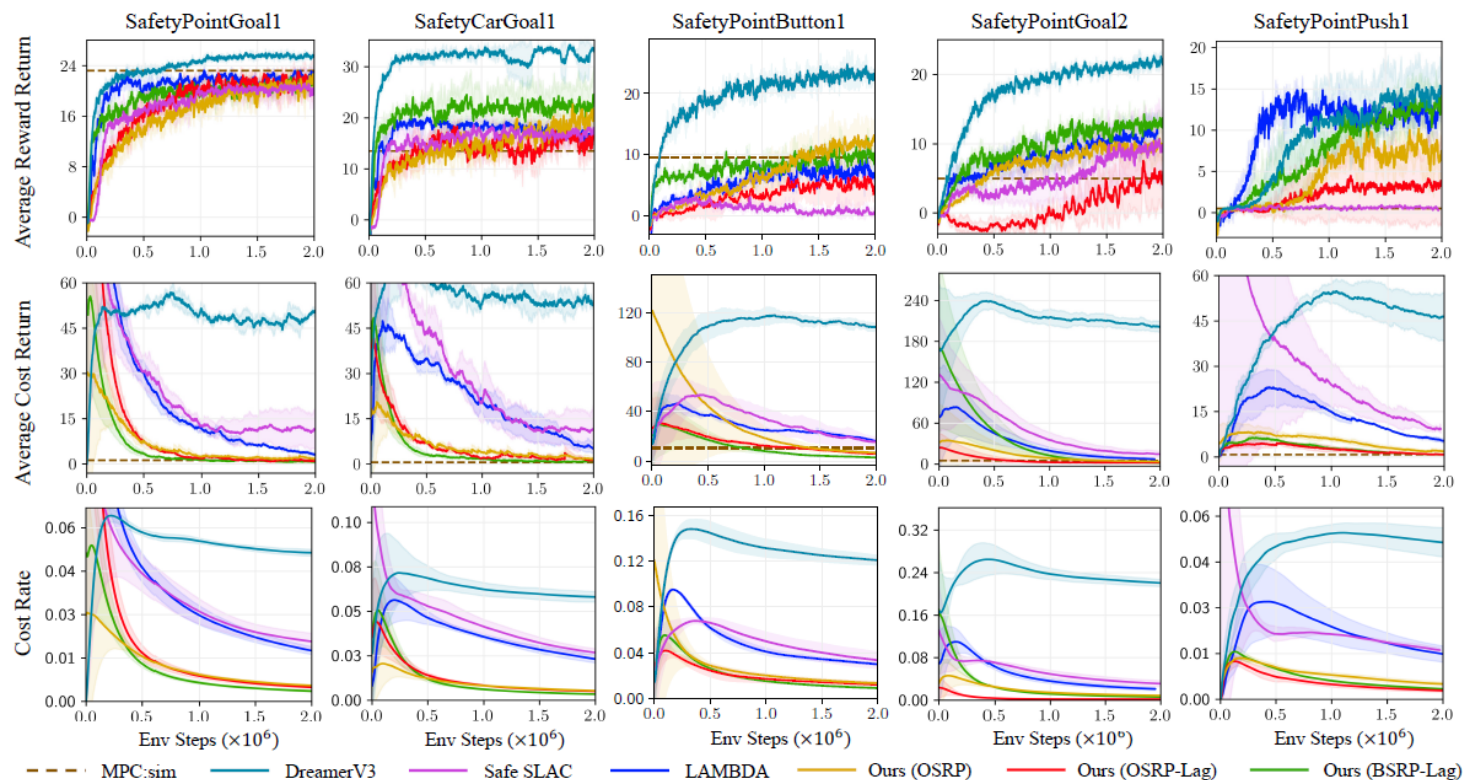


Figure 4: Comparing SafeDreamer to model-based baselines across five image-based safety tasks.

zero-cost performance

4. EXPERIMENTAL RESULTS

(2) Safety-Gymnasium tasks

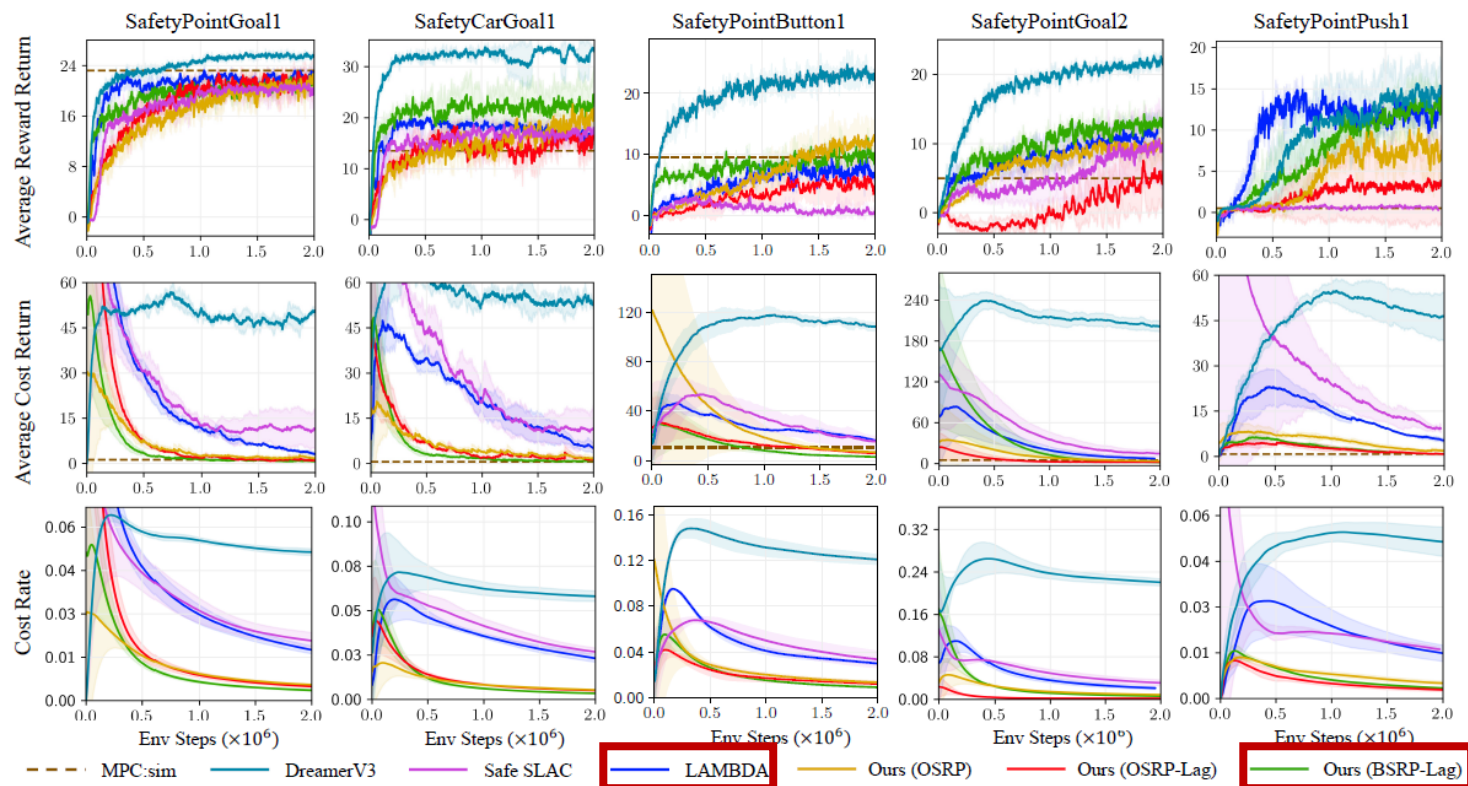


Figure 4: Comparing SafeDreamer to model-based baselines across five image-based safety tasks.

similar rewards, 94.3% reduction in costs

4. EXPERIMENTAL RESULTS

(2) Safety-Gymnasium tasks

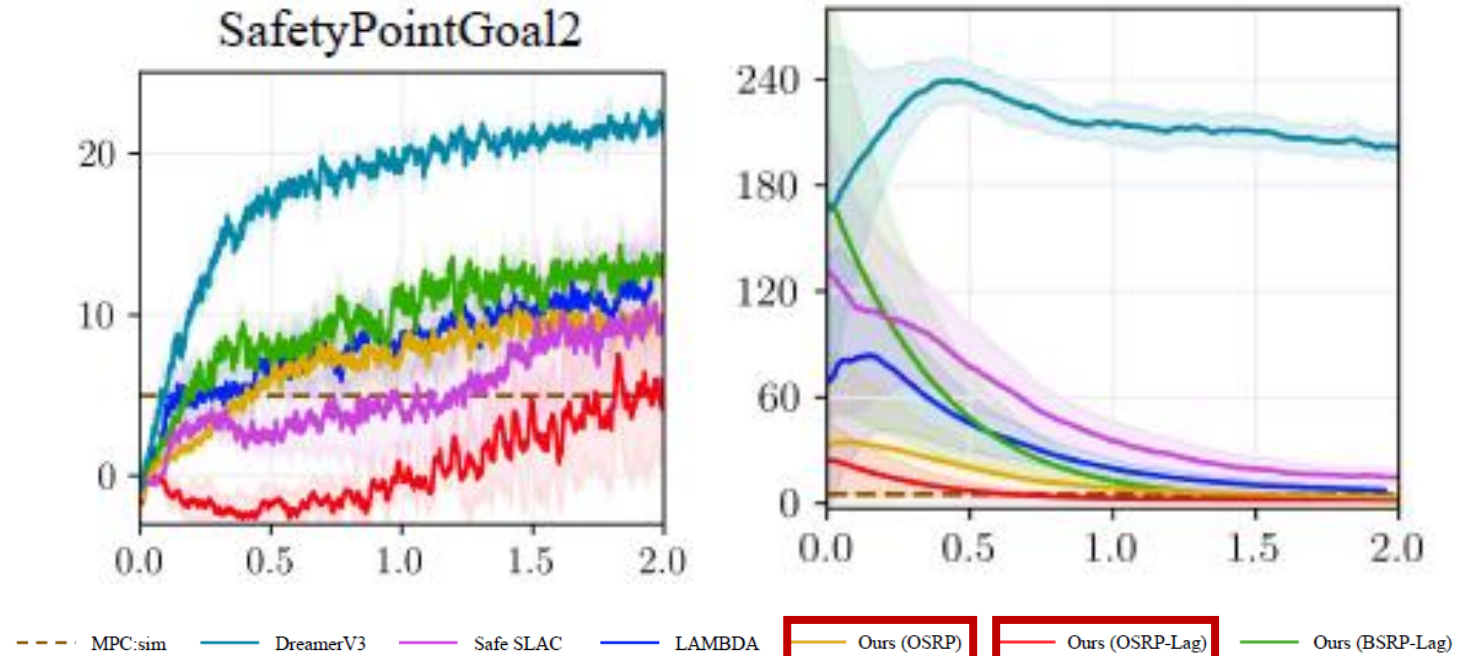


Figure 4: Comparing SafeDreamer to model-based baselines across five image-based safety tasks.

with more obstacles in the environment, OSRP-Lag tends to be more conservative than OSRP

4. EXPERIMENTAL RESULTS

(2) Safety-Gymnasium tasks

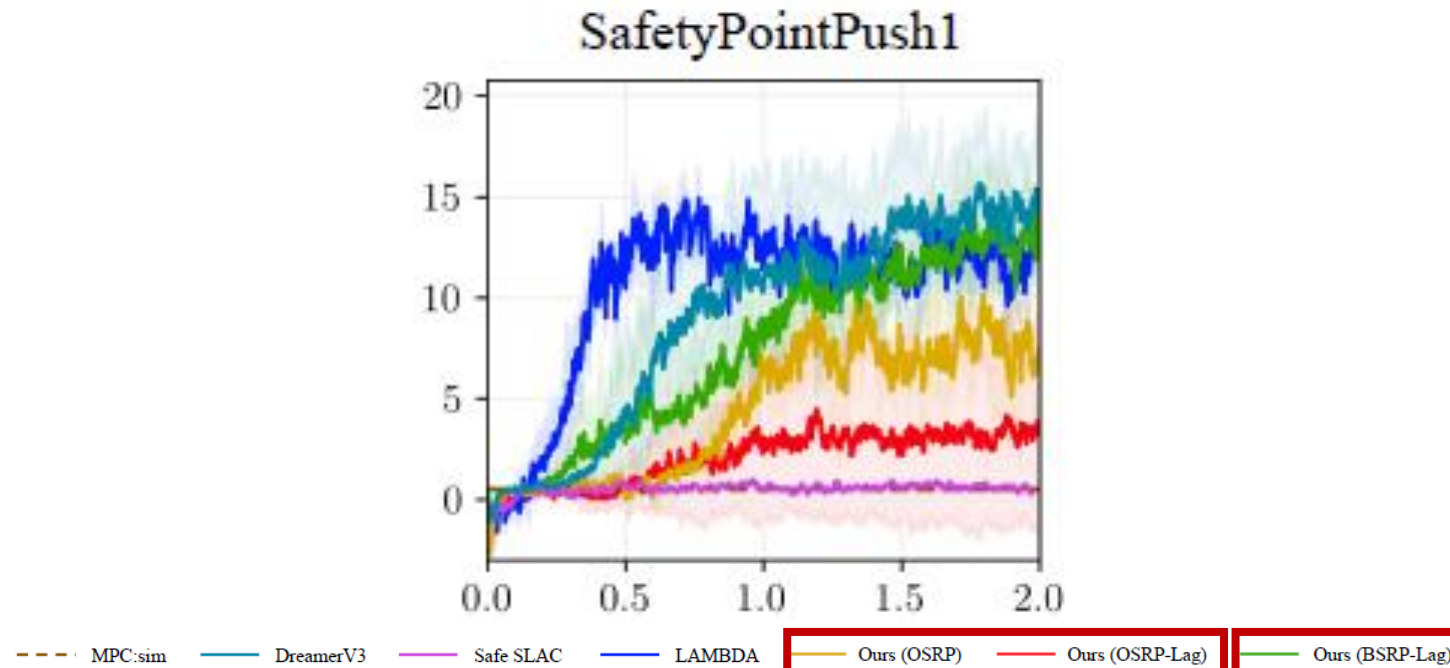


Figure 4: Comparing SafeDreamer to model-based baselines across five image-based safety tasks.

enable to predict dynamic environmental changes in real time!

limited length and number of planning trajectories (within a finite time)

4. EXPERIMENTAL RESULTS

(3) Low-dimensional tasks

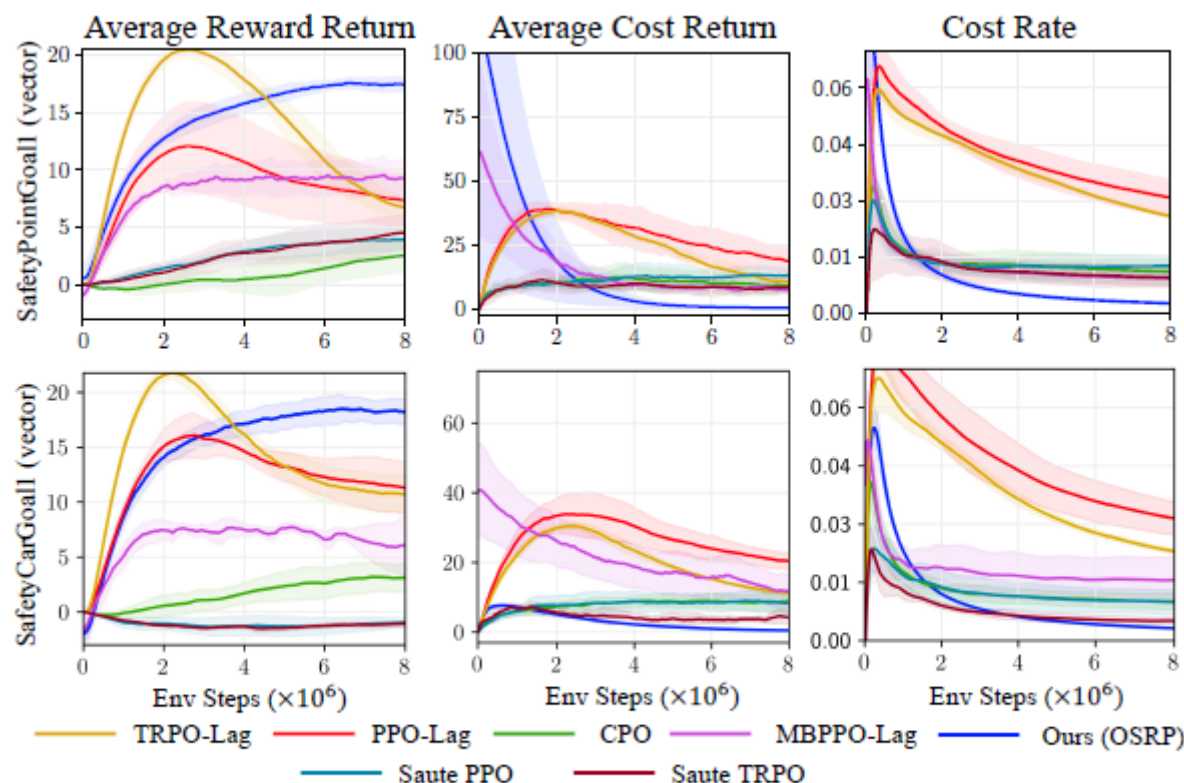


Figure 5: Results in low-dimensional input tasks.

SafeDreamer: Safe Reinforcement Learning with World Models

24.07.09

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Appendix

[Motivation]

- Safe SLAC(2022); **LAMBDA(2022)**; MBPPO-lag(2022) offer solutions to ensure cost and reward balance via the world model.
- Additionally, these methods fail to realize nearly zero-cost performance across several environments due to inaccuracies in modeling the agent's safety in current or future states.

Appendix

[Motivation]

- With the cost threshold nearing zero, existing Lagrangian-based methods often fail to meet constraints or meet them but cannot complete the tasks. Conversely, by employing an internal dynamics model, an agent can effectively plan action trajectories that secure high rewards and nearly zero costs (González et al., 2015).
- This underscores the significance of dynamics models and planning in such contexts. However, obtaining a ground-truth dynamics model for planning in complex scenarios like vision-only autonomous driving is impractical (Li et al., 2022). Additionally,
- the high expense of long-horizon planning forces optimization over a finite horizon, yielding locally optimal and unsafe solutions. To fill such gaps, we aim to answer the following question: *How can we develop a safety-aware world model to balance long-term rewards and costs of agent?*

Appendix

[Motivation]

- Autonomous Intelligence Architecture (LeCun, 2022) introduces a world model and incorporates costs that reflect the agent's level of discomfort. Subsequently, Hogewind et al. (2022); As et al. (2022); Jayant & Bhatnagar (2022) offer solutions to ensure cost and reward balance via the world model.
- However, these methods fail to realize nearly zero-cost performance across several environments due to inaccuracies in modeling the agent's safety in current or future states.
- In this work, we introduce SafeDreamer (see Figure 1a), which integrates safety planning and the Lagrangian method within a world model to balance errors between cost models and critics. A detailed comparison with various algorithms can be found in Table 1.

Appendix

[Model components]

Observation encoder: $z_t \sim E_\phi(z_t \mid h_t, o_t)$

Observation decoder: $\hat{o}_t \sim O_\phi(\hat{o}_t \mid s_t)$

Reward decoder: $\hat{r}_t \sim R_\phi(\hat{r}_t \mid s_t)$

Cost decoder: $\hat{c}_t \sim C_\phi(\hat{c}_t \mid s_t)$

Sequence model: $h_t, \hat{z}_t = S_\phi(h_{t-1}, z_{t-1}, a_{t-1})$

Actor: $a_t \sim \pi_\theta(a_t \mid s_t)$

Reward critic: $\hat{v}_{r_t} \sim V_{\psi_r}(\hat{v}_{r_t} \mid s_t)$

Cost critic: $\hat{v}_{c_t} \sim V_{\psi_c}(\hat{v}_{c_t} \mid s_t)$

Appendix

[Constrained cross-entropy method]

- The basic idea is to generate a sequence of policy distributions that eventually concentrates on a feasible (locally) optimal policy. Given a distribution over Π_Θ , we randomly generate a set of sample policies, sort them with a ranking function that depends on their G-values and H-values and then update the policy distribution with a subset of high-ranking sample policies.

Problem 1. Given a set $\Pi = \{\pi_\theta : \theta \in \Theta\}$ of policies with parameter space Θ , an NEF $F_\mathcal{V} = \{f_v \in \mathcal{D}(\Theta) : v \in \mathcal{V}\}$ of distributions over Θ , two functions $G : \Pi \rightarrow \mathbb{R}^+$ and $H : \Pi \rightarrow \mathbb{R}$, a constraint upper bound d and $\rho \in (0, 1)$, compute $v^* \in \mathcal{V}$ such that

$$v^* = \arg \max_{v \in \mathcal{V}} L(v; \rho),$$

where $L : \mathcal{V} \times (0, 1) \rightarrow \mathbb{R}$ is defined in (1) or (2).

Appendix

[Constrained cross-entropy method]

Definition 4.1. A parameterized family $F_{\mathcal{V}} = \{f_{\mathbf{v}} \in \mathcal{D}(\Theta), \mathbf{v} \in \mathcal{V} \subseteq \mathbb{R}^{d_{\mathbf{v}}}\}$ is called a *natural exponential family* if there exist continuous mappings $\Gamma : \mathbb{R}^{d_{\theta}} \rightarrow \mathbb{R}^{d_{\mathbf{v}}}$ and $K : \mathbb{R}^{d_{\theta}} \rightarrow \mathbb{R}$ such that $f_{\mathbf{v}}(\theta) = \exp(\mathbf{v}^{\top} \Gamma(\theta) - K(\mathbf{v}))$, where $\mathcal{V} \subseteq \{\mathbf{v} \in \mathbb{R}^{d_{\mathbf{v}}} : |K(\mathbf{v})| < \infty\}$ is the natural parameter space and $K(\mathbf{v}) = \log \int_{\Theta} \exp(\mathbf{v}^{\top} \Gamma(\theta)) d\theta$.

Appendix

[Constrained cross-entropy method]

Definition 4.2. [10] Given a distribution $P \in \mathcal{D}(\mathbb{R})$, $\rho \in (0, 1)$ and a random variable $X \sim P$, the ρ -quantile of X is defined as a scalar γ such that $Pr(X \leq \gamma) \geq \rho$ and $Pr(X \geq \gamma) \geq 1 - \rho$.

For $\rho \in (0, 1)$, $\mathbf{v} \in \mathcal{V}$ and any function $X : \Theta \rightarrow \mathbb{R}$, we denote the ρ -quantile of X for $\theta \sim f_{\mathbf{v}}$ by $\xi_X(\rho, \mathbf{v})$. We also define $\delta : \mathbb{R} \times \{\geq, \leq, >, <, =\} \times \mathbb{R} \rightarrow \{0, 1\}$ as an indicator function such that for $\circ \in \{\geq, \leq, >, <, =\}$, $\delta(x \circ y) = 1$ if and only if $x \circ y$ holds. The surrogate objective function for the unconstrained CE method is $\mathbb{E}_{\theta \sim f_{\mathbf{v}}}[G(\pi_{\theta})\delta(G(\pi_{\theta}) \geq \xi_G(1 - \rho, \mathbf{v}))]$. In other words, a policy π_{θ} is considered as highly ranked if $G(\pi_{\theta}) \geq \xi_G(1 - \rho, \mathbf{v})$. When there is a constraint $H(\pi) \leq d$, we define $U : \Pi_{\Theta} \rightarrow \mathbb{R}$ such that $U(\pi_{\theta}) := G(\pi_{\theta})\delta(H(\pi_{\theta}) \leq d)$ for any $\theta \in \Theta$ and extend the surrogate function as follows:

$$L(\mathbf{v}; \rho) := \begin{cases} \mathbb{E}_{\theta \sim f_{\mathbf{v}}}[G(\pi_{\theta})\delta(H(\pi_{\theta}) \leq \xi_H(\rho, \mathbf{v}))], & \text{if } \xi_H(\rho, \mathbf{v}) > d; \\ \mathbb{E}_{\theta \sim f_{\mathbf{v}}}[U(\pi_{\theta})\delta(U(\pi_{\theta}) \geq \xi_U(1 - \rho, \mathbf{v}))], & \text{otherwise.} \end{cases} \quad (1)$$

We can combine the two cases. Define $S : \Pi_{\Theta} \times \mathcal{V} \times (0, 1) \rightarrow \{0, 1\}$ such that

$$S(\pi_{\theta}, \mathbf{v}, \rho) := \delta(\xi_H(\rho, \mathbf{v}) > d)\delta(H(\pi_{\theta}) \leq \xi_H(\rho, \mathbf{v})) + \\ \delta(\xi_H(\rho, \mathbf{v}) \leq d)\delta(H(\pi_{\theta}) \leq d)\delta(U(\pi_{\theta}) \geq \xi_U(1 - \rho, \mathbf{v})),$$

then (1) can be rewritten as

$$L(\mathbf{v}; \rho) = \mathbb{E}_{\theta \sim f_{\mathbf{v}}}[G(\pi_{\theta})S(\pi_{\theta}, \mathbf{v}, \rho)]. \quad (2)$$

Appendix

[Constrained cross-entropy method]

Algorithm 1 Constrained Cross-Entropy Method

Require: An objective function G , a constraint function H , a constraint upper bound d , a class of parameterized policies Π_{Θ} , an NEF family $F_{\mathcal{V}}$.

- 1: $l \leftarrow 1$. Initialize $n_l, \mathbf{v}_l, \rho, \lambda_l, \alpha_l$. $k_l \leftarrow \lceil \rho n_l \rceil$. $\hat{\eta}_l \leftarrow \mathbf{0}$.
 - 2: **repeat**
 - 3: Sample $\theta_1, \dots, \theta_{n_l} \sim f_{\mathbf{v}_l}$ i.i.d..
 - 4: **for** $i = 1, \dots, n_l$ **do**
 - 5: Simulate π_{θ_i} and estimate $G(\pi_{\theta_i}), H(\pi_{\theta_i})$.
 - 6: **end for**
 - 7: Sort $\{\theta_i\}_{i=1}^{n_l}$ in ascending order of H . Let Λ_l be the first k_l elements.
 - 8: **if** $H(\pi_{\theta_{k_l}}) \leq d$ **then**
 - 9: Sort $\{\theta_i \mid H(\pi_{\theta_i}) \geq d\}$ in descending order of G . Let Λ_l be the first k_l elements.
 - 10: **end if**
 - 11: $\hat{\eta}_{l+1} \leftarrow \alpha_l \sum_{\theta \in \Lambda_l} \frac{G(\pi_{\theta})}{\sum_{\theta \in \Lambda_l} G(\pi_{\theta})} \Gamma(\theta) + (1 - \alpha_l) \left(\frac{\lambda_l}{n_l} \sum_{i=1}^{n_l} \Gamma(\theta_i) + (1 - \lambda_l) \hat{\eta}_l \right)$.
 - 12: $\mathbf{v}_{l+1} \leftarrow m^{-1}(\hat{\eta}_{l+1})$.
 - 13: Update n_l, λ_l, α_l . $l \leftarrow l + 1$. $k_l \leftarrow \lceil \rho n_l \rceil$.
 - 14: **until** Stopping rule is satisfied.
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Appendix

[Constrained cross-entropy method]

- The basic idea is to generate a sequence of policy distributions that eventually concentrates on a feasible (locally) optimal policy. Given a distribution over Π_Θ , we randomly generate a set of sample policies, sort them with a ranking function that depends on their G-values and H-values and then update the policy distribution with a subset of high-ranking sample policies.

Problem 1. Given a set $\Pi = \{\pi_\theta : \theta \in \Theta\}$ of policies with parameter space Θ , an NEF $F_\mathcal{V} = \{f_v \in \mathcal{D}(\Theta) : v \in \mathcal{V}\}$ of distributions over Θ , two functions $G : \Pi \rightarrow \mathbb{R}^+$ and $H : \Pi \rightarrow \mathbb{R}$, a constraint upper bound d and $\rho \in (0, 1)$, compute $v^* \in \mathcal{V}$ such that

$$v^* = \arg \max_{v \in \mathcal{V}} L(v; \rho),$$

where $L : \mathcal{V} \times (0, 1) \rightarrow \mathbb{R}$ is defined in (1) or (2).